2025.01.15 / Repetitorium

Dienstag, 14. Jänner 2025 20:24

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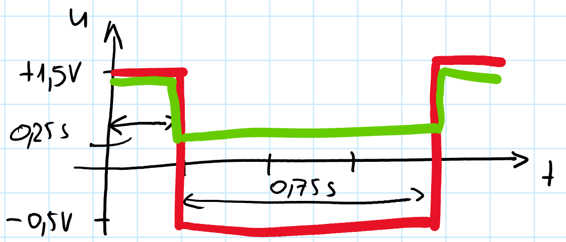
ZEITABHÄNGIGKEIT



Periodisches Signal

$T = 1\,s \qquad f = 1\,Hz$

$\omega = 2\pi f = 2\pi\, s^{-1}$

$U_{Max} = +1{,}5V \qquad U_{Min} = -0{,}5V \qquad \hat{U} = 1{,}5V \qquad U_{pp} = 2V$

$$\bar{u} = \frac{1}{T}\int_0^T u\,dt = \frac{1}{1s}\left(\int_0^{0{,}25} 1{,}5V\,dt + \int_{0{,}25}^{1{,}0} -0{,}5\,dt\right) =$$

$$= \frac{1}{1s}\left(\left[1{,}5t\right]_0^{0{,}25} + \left[-0{,}5t\right]_{0{,}25}^{1{,}0}\right) = \frac{1}{1s}\left(0{,}375 - 0 - 0{,}5 + 0{,}125\right)Vs =$$

$$= \frac{1}{1s}\,0\,Vs = \underline{0V}$$

$$|\bar{u}| = \frac{1}{T}\int_T |u|\,dt = \frac{1}{1s}\left(\int_0^{0{,}25} 1{,}5V\,dt + \int_{0{,}25}^{1{,}0} 0{,}5V\,dt\right) =$$

$$= \frac{1}{1s}\left(0{,}375 - 0 + 0{,}5 - 0{,}125\right)Vs = 0{,}75V$$

$$U = \sqrt{\frac{1}{T}\int_0^T u^2\,dt}$$

$$\int_0^T u^2\,dt = \int_0^{0{,}25} 1{,}5^2 V^2\,dt + \int_{0{,}25}^{1{,}0} 0{,}5^2 V^2\,dt =$$

$$= 2{,}25V^2 t\Big|_0^{0{,}25} + 0{,}25V^2 t\Big|_{0{,}25}^{1{,}0} =$$

$$= (2{,}25 \cdot 0{,}25 - 0 + 0{,}25 \cdot 1 - 0{,}0625)\,V^2 s$$

$$= 0{,}75\,V^2 s$$

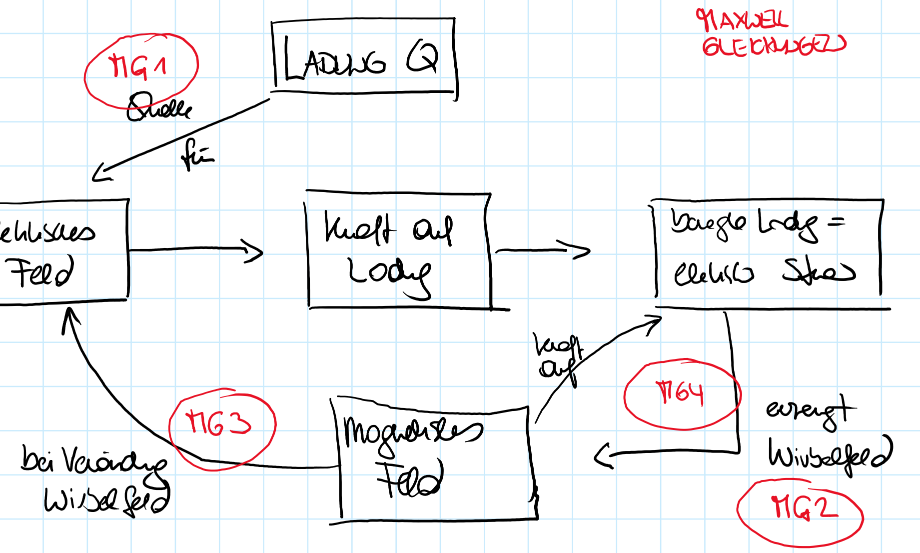
$$\underline{U} = \sqrt{\frac{1}{1s}\,0{,}75\,V^2 s} = \underline{0{,}866\,V}$$

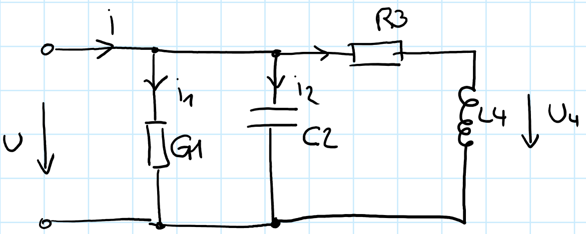
Sc[illegible] $= \frac{\hat{u}}{[illegible]} = \frac{1{,}5V}{[illegible]} = 1{,}7$[illegible]

$$\text{Scheitelfaktor} = \frac{\hat{u}}{U} = \frac{1{,}5\,V}{0{,}866\,V} = 1{,}732$$

$$\text{Formfaktor} = \frac{U}{|\bar{u}|} = \frac{0{,}866\,V}{0{,}75\,V} = 1{,}154$$

WECHSELSTROMTECHNIK





$$u = 5V \cdot \cos(6283\,s^{-1} t)$$

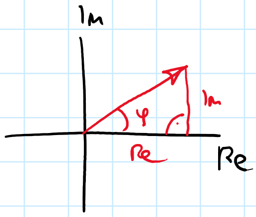
$G_1 = 1\,mS$ $\quad C_2 = 120\,nF$ $\quad R_3 = 1{,}5\,k\Omega$ $\quad L_4 = 159\,mH$

a) $\omega = 2\pi f = 6283\,s^{-1} \rightarrow f = 1\,kHz$

$\hat{u} = 5V$ $\quad \varphi_u = 0°$ $\quad U = \frac{5}{\sqrt{2}}V = 3{,}54\,V$

$\underline{U} = 3{,}54\,V \angle 0°$

b) $\underline{Y}$ $\quad \underline{Z}$



$$\underline{Z}_{34} = R_3 + j\omega L_4 = 1500\,\Omega + j\,999\,\Omega$$

$$\underline{Y}_{34} = \frac{1}{\underline{Z}_{34}} = \frac{1}{1500\,\Omega + j\,999\,\Omega} = \frac{1500 - j999}{1500^2 + 999^2} \cdot \frac{1}{\Omega} =$$

$$= 0{,}461\,mS - j\,0{,}308\,mS = 0{,}55\,mS \angle -33{,}7°$$

$$\underline{Y} = G_1 + j\omega C_2 + \underline{Y}_{34} = 1\,mS + j\,0{,}754\,mS + 0{,}461\,mS - j0{,}308\,mS =$$

$$= 1{,}461\,mS + j\,0{,}446\,mS = 1{,}528\,mS \angle 17°$$

$$\underline{Z} = \frac{1}{\underline{Y}} = 654{,}6\,\Omega \angle -17°$$

c) $$\underline{I}_2 = \frac{\underline{U}}{\underline{X}_{C2}} = j\omega C2 \cdot \underline{U} = 0{,}754\,mS \angle 90° \cdot 3{,}54\,V \angle 0° = 2{,}67\,mA \angle 90°$$

$$i_2(t) = 3{,}77\,mA \cos(6283\,s^{-1} t + 90°)$$

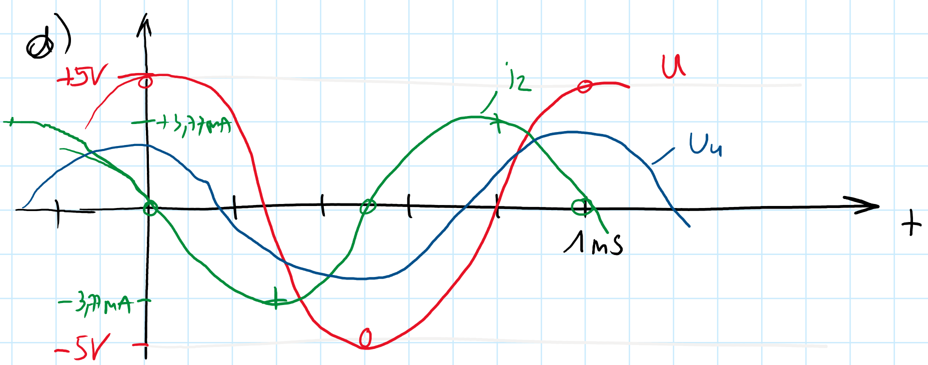
$$\underline{U}_4 = \frac{j\omega L_4}{\underline{Z}_{34}} \underline{U} = \frac{j\,999\,\Omega}{1500\,\Omega + j\,999\,\Omega} \, 3{,}54\,V \angle 0° =$$

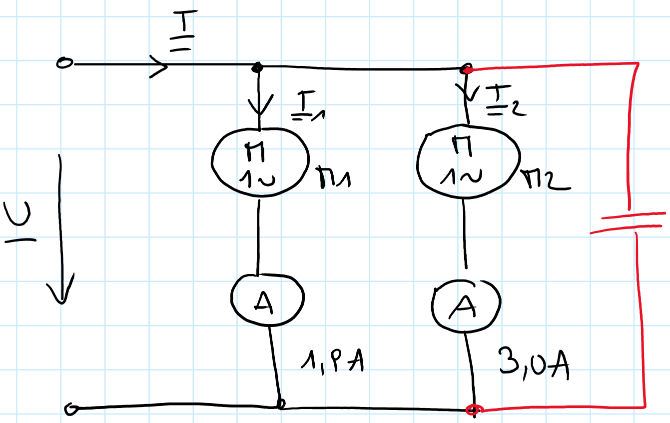
$$= (0{,}307 + j\,0{,}461) \cdot 3{,}54\,V =$$

$$= 1{,}087\,V + j\,1{,}632\,V = 1{,}961\,V \angle 56{,}3°$$

$$u_4 = 2{,}773\,V \cos(6283\,s^{-1} t + 56{,}3°)$$

d)

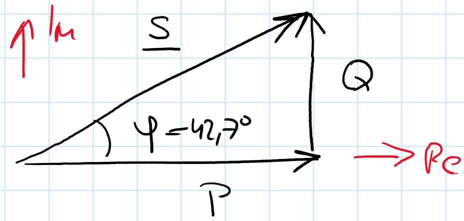




$U = 230\,V$ $f = 50\,Hz$

$P_M = 400\,W$

M1 teilkompensiert



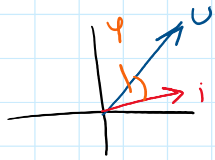
a) berechne $\underline{I}$ sowie P, Q, S

$S_1 = U \cdot I_1 = 437\,VA$ $S_2 = U \cdot I_2 = 690\,VA$

$\cos\varphi_1 = \frac{P}{S_1} = \frac{400}{437} = 0{,}915$ $\cos\varphi_2 = \frac{P_2}{S_2} = \frac{400}{690} = 0{,}58$

$\varphi_1 = 23{,}8°$ $\varphi_2 = 54{,}6°$

$\varphi_1 = 23{,}8^\circ$ $\qquad$ $\varphi_2 = 54{,}6^\circ$



$\varphi = \varphi_u - \varphi_i$

Annahme $\quad \underline{U} = 230\,V \angle 0^\circ$

$\varphi_{i1} = -23{,}8^\circ$ $\qquad$ $\varphi_{i2} = -54{,}6^\circ$

$\underline{I}_1 = 1{,}9\,A \angle -23{,}8^\circ$ $\qquad$ $\underline{I}_2 = 3{,}0\,A \angle -54{,}6^\circ$

$\underline{I} = \underline{I}_1 + \underline{I}_2 = 4{,}73\,A \angle -42{,}7^\circ$

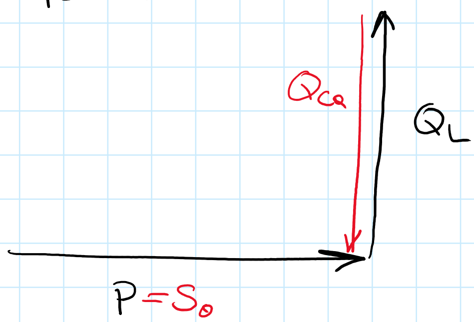
$\underline{S} = \underline{U}\,\underline{I}^* = 230\,V \angle 0^\circ \cdot 4{,}73\,A \angle 42{,}7^\circ = 1.088\,VA \angle 42{,}7^\circ =$

$= 800\,W + j\,738\,var = P + jQ$

b) Kompensation $\quad \cos\varphi_a = 1 \qquad \cos\varphi_b = 0{,}95$

$Q_{Ca} = -Q_L = -738\,var = -\omega C_a U^2$

$C_a = -\dfrac{Q_{Ca}}{\omega U^2} = 44{,}4\,\mu F$



$\cos\varphi_b = 0{,}95 \quad + \quad P = \text{konst.}$

$\varphi_b = 18{,}2^\circ$

$Q_{Lb} = P \cdot \tan\varphi_b = 263\,var$

$Q_{Cb} = -(Q_L - Q_{Lb}) = -475\,var$

$C_b = -\dfrac{Q_{Cb}}{\omega U^2} = 28{,}6\,\mu F$